UNIT 1 MINI PROJECT

ENTERPRISE STUDENT RECORD AND ACADEMIC MANAGEMENT PLATFORM

Product Engineering • Design Thinking • Requirements • Architecture

Table of Contents

1. 1. Executive Summary
2. 2. Introduction to Software Product Development
3. 3. Business Understanding: Higher Education Operations
4. 4. Business Requirements Document (BRD)
5. 5. Product Vision and Business Objectives
6. 6. Stakeholder Analysis
7. 7. User Personas
8. 8. Requirements Engineering – FRS
9. 9. Requirements Engineering – NFRS
10. 10. Design Thinking Report
11. 11. IBM Design Thinking
12. 12. IBM Observe → Reflect → Make Loop
13. 13. Minimum Viable Product (MVP)
14. 14. Product Roadmap
15. 15. Initial System Architecture
16. 16. Project Charter
17. 17. Repository and Git Structure
18. 18. Product Discovery Documentation
19. 19. Risk, Quality and Success Measures
20. 20. Conclusion

1. Executive Summary

The Enterprise Student Record and Academic Management Platform is a proposed unified digital system for higher-education academic operations. It is designed to connect student onboarding, course enrollment, course registration, attendance, examination, grading, result publishing and academic reporting in a centralized environment.

The project follows a product-engineering mindset: understand the institution and its users first, define measurable outcomes, design iteratively, build a focused MVP, and establish an architecture and delivery roadmap that can evolve after the first release.

The core value proposition is simple: reduce manual administrative effort, improve academic-record accuracy, provide timely information to students and faculty, and give administrators centralized visibility.

1.1 Project at a Glance

Aspect	Definition
Product	Enterprise Student Record and Academic Management Platform
Primary Users	Students, Faculty, Academic Administrators, Examination Cell, IT Support
Core Problem	Fragmented/manual academic records and slow information flow
MVP	Login/RBAC, profiles & courses, attendance, marks/SGPA, reports
Design Approach	Design Thinking + IBM Design Thinking
Engineering Direction	Layered application architecture with database, testing, CI/CD and future cloud deployment

2. Introduction to Software Product Development

Modern software products are developed through structured engineering frameworks rather than ad-hoc programming. Coding is one execution phase inside a larger lifecycle that includes research, requirements, architecture, user experience, quality assurance, deployment and continuous monitoring.

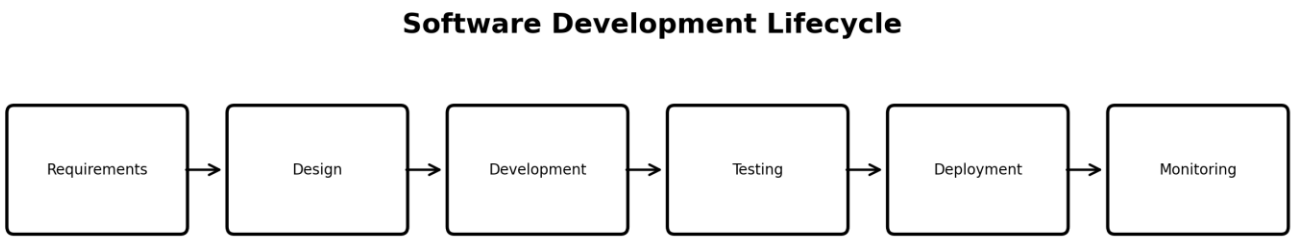
2.1 Programming vs Software Engineering

Dimension	Programming / Coding	Software Product Engineering
Primary Goal	Translate logic and algorithms into executable code.	Build scalable, maintainable and valuable systems for end users.
Scope	Functions, scripts or isolated tasks.	Full lifecycle: SDLC, architecture, UX, database, security, testing and delivery.
Success Metric	Code compiles and passes basic test inputs.	User adoption, reliability, performance, maintainability and business outcomes.
Mindset	Technology-centric: How do I write this function?	Product-centric: What problem are we solving and what outcome should users achieve?

2.2 Root Causes of Software Project Failure

- Misaligned requirements: building features users do not need or cannot use effectively.
- Technology-centric bias: choosing tools before understanding the domain problem.
- Communication silos between business, development, QA and operations teams.
- Lack of iterative feedback: waiting until late development to discover usability or workflow problems.
- Uncontrolled scope: adding features faster than the team can validate and deliver them.

2.3 SDLC Workflow



The SDLC provides a repeatable path from requirements to operation. In an agile product, the lifecycle is not strictly one-directional: feedback can return the team to requirements or design so the product can improve.

3. Business Understanding: Higher Education Operations

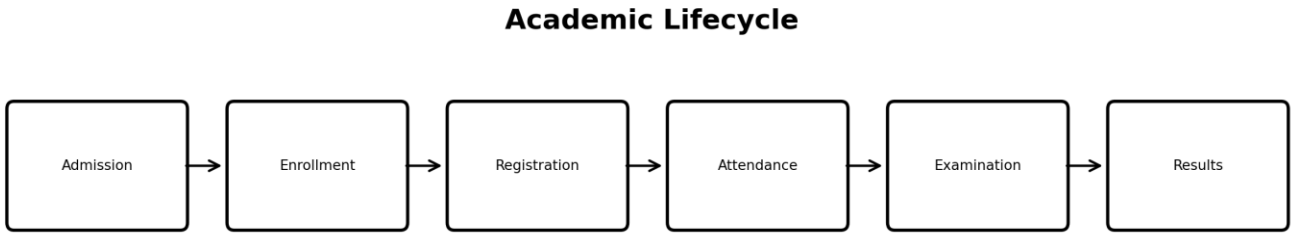
Before designing technical components, the engineering team must understand how academic information is created, changed, verified and consumed.

3.1 Core Academic Operations

- Student onboarding and profile management: admission, permanent registration number (PRN), degree path and profile data.
- Course enrollment and curriculum mapping: semester offerings, electives, credits, prerequisites and faculty allocation.
- Attendance tracking: daily lecture records, percentage calculation, shortage warnings and leave-related workflows.
- Examination and grading: continuous assessment, end-semester marks, GPA/CGPA calculation, result publishing and revaluation workflows.
- Administrative and compliance reporting: consolidated grade sheets, transcript verification, audit records and institutional reports.

3.2 Manual Bottlenecks and Automated Opportunities

Manual Bottleneck	Platform Solution	Expected Business Outcome
Paper attendance registers	Mobile/web attendance logging	Shorter attendance entry time and easier shortage tracking.
Manual GPA/CGPA calculation	Configurable grading engine	Faster and more consistent result generation.
Physical result/transcript distribution	Centralized self-service portal	Faster access to academic information.
Siloed spreadsheets and files	Centralized relational database + audit trail	Better data consistency and traceability.



4. Business Requirements Document (BRD)

4.1 Business Problem Statement

Students and administrative personnel can experience delays in course enrollment, attendance tracking, mark publishing and transcript validation when academic information is maintained through fragmented manual or disconnected processes. Faculty and examination staff may also lack centralized visibility, increasing administrative effort and the chance of inconsistent records.

4.2 Business Justification

A unified platform can centralize academic information, reduce repetitive data entry, improve transparency, accelerate reporting and create a consistent source of academic truth. The product therefore addresses both operational efficiency and the quality of the student experience.

4.3 Business Objectives

21. Centralize academic records across the major student lifecycle stages.
22. Reduce manual administrative effort and repetitive record handling.
23. Improve the accuracy and consistency of academic information.
24. Provide timely access to attendance, course and result information.
25. Enable administrators and examination staff to generate reliable reports.
26. Create a foundation for future analytics and digital academic services.

4.4 Scope

In Scope	Deferred / Future Scope
Student profiles and admission records	Hostel management
Course enrollment and registration	Fee-payment gateway integration
Attendance management	Alumni portal
Examination and result processing	AI-based performance prediction
Academic reports and transcripts	Native mobile application

4.5 Assumptions and Constraints

- Users have access to basic computing devices and network connectivity.
- Institutional workflows and grading rules can be documented and configured.
- Access to student records is controlled by role and institutional policy.
- The first release may be piloted with a department, batch or limited user group.

5. Product Vision and Business Objectives

5.1 Product Vision Statement

To deliver a unified, secure and intuitive Student Record and Academic Management Platform that eliminates avoidable administrative overhead, accelerates academic information flow and empowers students, faculty and administrators with reliable access to educational records and metrics.

5.2 Mission Statement

Digitize and streamline the academic lifecycle from onboarding to result declaration through a reliable platform that reduces manual work while improving transparency, consistency and user experience.

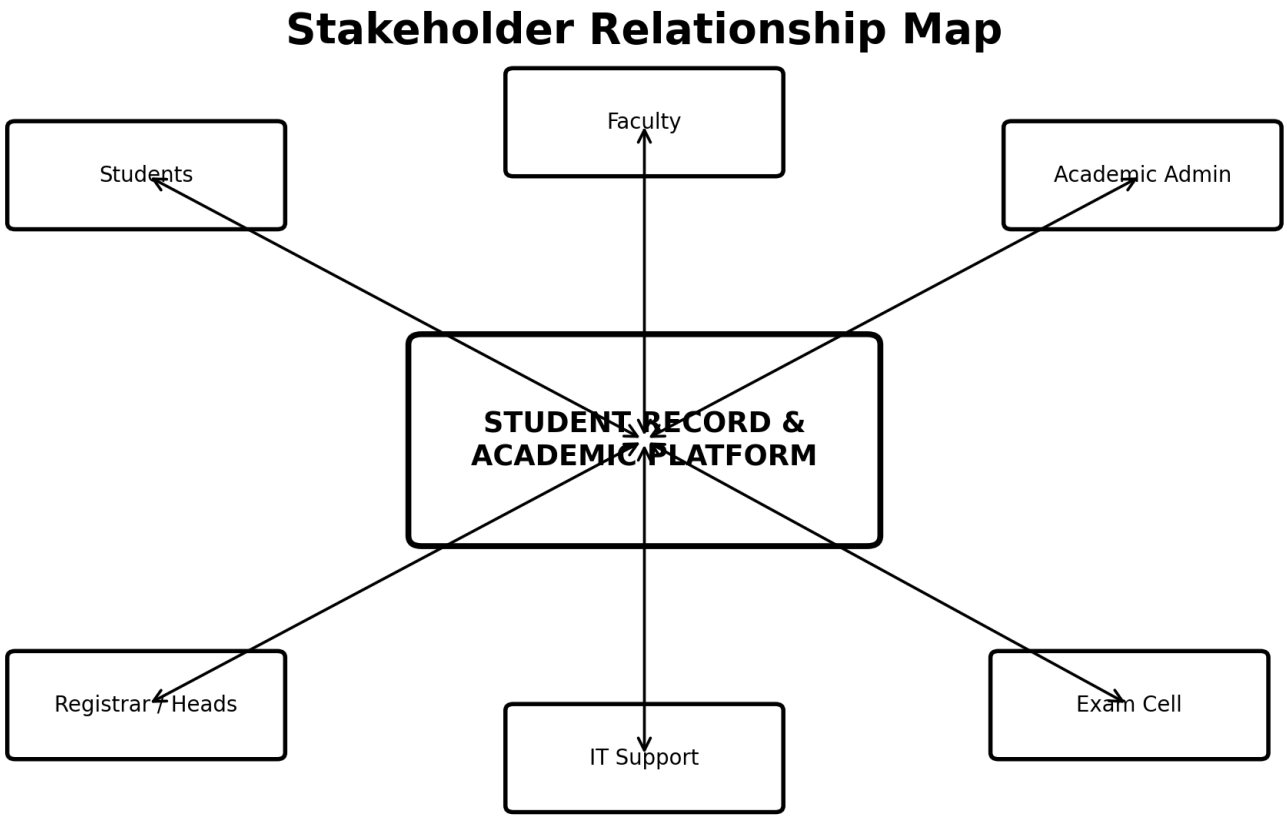
5.3 User Outcomes

User	Desired Outcome
Student	Complete registration and access attendance/results without unnecessary queues or delays.
Faculty	Record attendance and marks efficiently without duplicate manual entry.
Academic Administrator	Monitor academic records and generate reliable reports from one system.
Examination Cell	Process marks and publish verified results efficiently and securely.
IT Support	Maintain access, reliability and controlled system operation.

5.4 Success Criteria

- High accuracy and consistency of academic records.
- Fast access to common student and academic queries.
- Reduced administrative turnaround time.
- Positive usability feedback from pilot users.
- Traceable access and change history for sensitive academic operations.

6. Stakeholder Analysis



Stakeholder Group	Key Roles	Responsibilities / Expectations
Students	Primary end users	Registration, attendance visibility, results and notifications.
Faculty	Academic operators	Attendance, marks, course-related updates and student progress.
Academic Administrators	Business owners / coordinators	Workflow definition, reporting, policy and academic monitoring.
Examination Cell	Result-processing stakeholders	Secure marks processing, result publication and transcript workflows.
IT Support	Technical operations	Accounts, infrastructure, access control, troubleshooting and maintenance.
Department Heads / Registrar Office	Governance stakeholders	Institutional oversight, compliance and decision-making.

6.1 Stakeholder Priority Matrix

Stakeholder	Influence	Interest	Engagement Approach
Academic Administrators	High	High	Frequent review and workflow validation
Examination Cell	High	High	Detailed requirements and acceptance testing
Faculty	Medium	High	Usability testing and iterative feedback
Students	Medium	High	Interviews, prototype testing and pilot feedback

IT Support	High	Medium	Architecture, security and operational reviews

7. User Personas

Persona	Primary Objectives	Pain Points	Target Features
Riya – Student	Register courses, track attendance and view results.	Delayed information, limited attendance transparency, deadline uncertainty.	Responsive dashboard, attendance alerts, result visibility, notifications.
Prof. Deshmukh – Faculty	Record daily attendance and enter evaluation marks efficiently.	Paper registers, repetitive entry, time pressure during exams.	Quick attendance entry, bulk marks entry, shortage tracker.
Dr. Kulkarni – Examination Administrator	Process results securely and generate transcripts/reports.	Manual mark-entry risk and long processing cycles.	Secure grading workflow, GPA/CGPA engine, transcript export.

7.1 Persona Design Principle

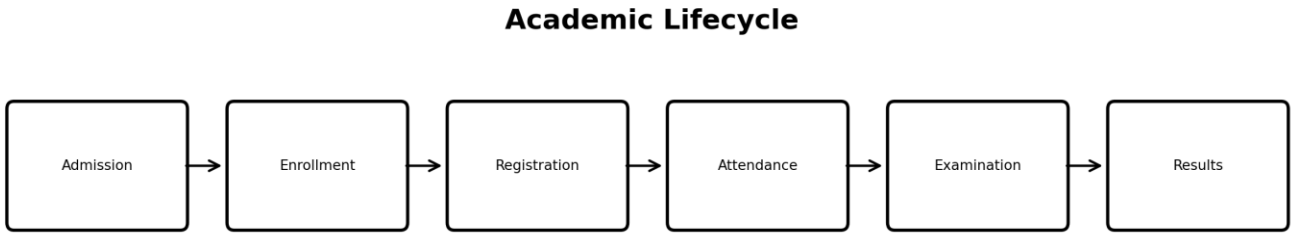
Each feature should be connected to a real user outcome. A dashboard is valuable not because it is visually impressive, but because it helps a student or administrator complete a task faster and with fewer errors.

8. Requirements Engineering – Functional Requirements Specification (FRS)

Functional requirements define what the system must do.

ID	Requirement	Acceptance-Oriented Description
FR-01	Authentication & RBAC	Secure login with role-based permissions for Student, Faculty, Academic Admin, Examination Cell and IT Support.
FR-02	Student Profile & Admission	Maintain student profile, PRN, admission information and academic mapping.
FR-03	Course & Enrollment	Allow authorized users to manage active semester courses, electives, credits and registration.
FR-04	Attendance Management	Faculty can record attendance; system calculates percentages and identifies low-attendance cases.
FR-05	Examination & Results	Capture marks, apply configured grading rules, calculate SGPA/CGPA and prepare result reports.
FR-06	Reporting	Generate academic and administrative reports with controlled access.
FR-07	Notifications	Send or display alerts for registration deadlines, attendance shortages and result publication.

8.1 Functional Workflow



8.2 Example User Story

As a student, I want to see my current attendance percentage by course so that I can identify shortages before examinations and take appropriate action.

Acceptance idea: after authentication, the student can open the attendance view, see course-wise percentages, identify any shortage indicator, and access the latest recorded session information permitted by the system.

9. Requirements Engineering – Non-Functional Requirements (NFRS)

Non-functional requirements define system qualities and measurable constraints. Where the project guide provides target values, these are treated as design targets rather than claims about an already deployed system.

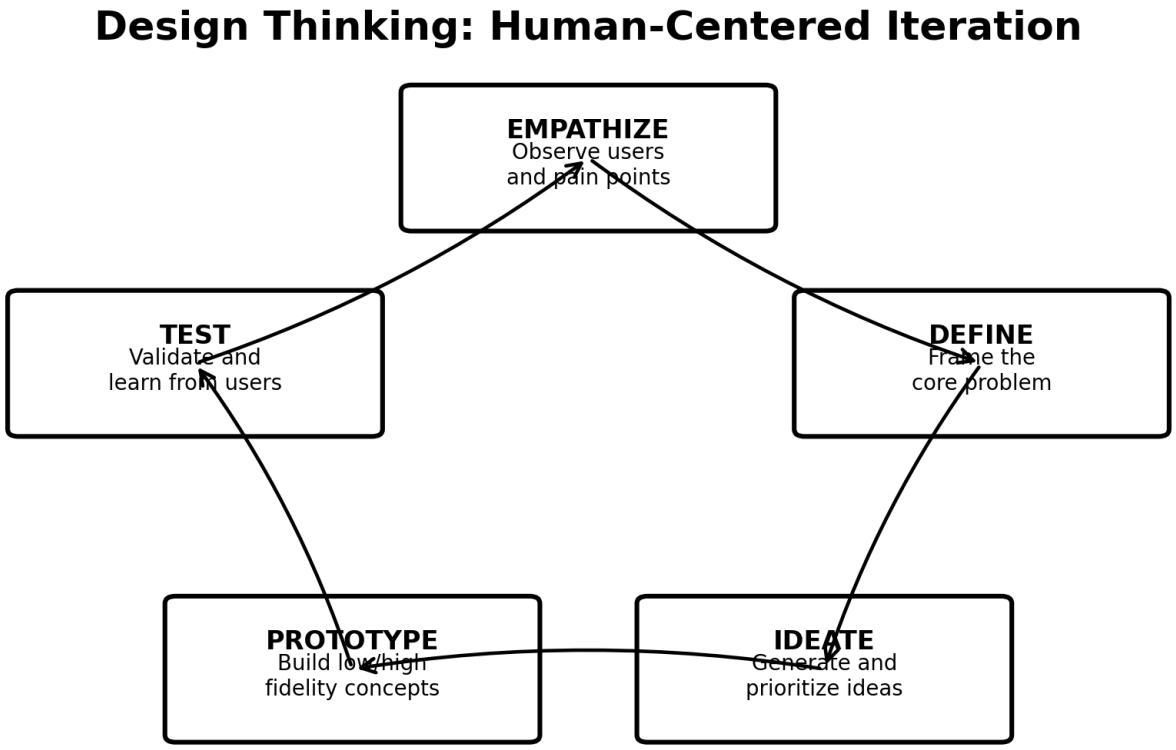
ID	Quality Area	Target / Requirement
NFR-01	Security	Secure password hashing such as Bcrypt/Argon2; secure authentication tokens; HTTPS; strict RBAC isolation.
NFR-02	Performance	Target sub-500 ms response for common record lookups and mark submissions under defined peak conditions.
NFR-03	Availability & Scalability	Design toward 99.9% availability and scalable deployment for peak examination/result traffic.
NFR-04	Usability & Accessibility	Responsive interface with accessibility considered against WCAG 2.1 AA principles.
NFR-05	Maintainability	Modular separation of admissions, registration, attendance, examination and reporting components.
NFR-06	Auditability	Sensitive academic actions should record actor, timestamp and relevant change information.
NFR-07	Backup & Recovery	Academic records should have scheduled backup and a documented recovery approach.

9.1 Security-by-Design Principles

- Least privilege: users receive only the permissions required for their role.
- Defense in depth: authentication, authorization, transport security and database controls work together.
- Input validation: user-provided data is validated before processing or storage.
- Auditability: sensitive operations are traceable without exposing unnecessary personal information.

10. Design Thinking Report

Design Thinking is an iterative, human-centered problem-solving approach. The purpose is to understand user needs before committing to a technical solution.



10.1 Empathize

Observe students, faculty, academic administrators and examination staff while they perform common academic tasks. Record pain points such as repeated data entry, waiting, unclear status information, attendance uncertainty and result-processing delays.

10.2 Define

Core problem statement: Students and academic personnel experience delays and errors in managing enrollment, attendance, marks and academic records because information can be fragmented across manual or disconnected processes. Faculty and examination staff also need better centralized visibility.

10.3 Ideate

- Automated low-attendance notifications.
- Digital student identity/profile access.
- Faculty attendance and gradebook interface.
- Real-time academic progress dashboard.
- Dynamic transcript and report generation.

10.4 Prototype

Prototype the Student Dashboard, Faculty Attendance Page, Examination Grading Panel and Administrator Analytics Dashboard. Start with low-fidelity wireframes so navigation and workflow can be tested before expensive implementation.

10.5 Test

Test prototypes with representative students and staff. Observe task completion, confusion points, accessibility issues and missing information. Feed the findings back into the next prototype.

11. IBM Design Thinking

IBM Design Thinking extends human-centered design for enterprise environments by emphasizing measurable outcomes, continuous reinvention and cross-functional collaboration.

11.1 Focus on User Outcomes

A technical feature should be connected to an outcome. For example, instead of only defining a database query endpoint, the product goal can be framed as enabling an authorized academic advisor to retrieve a student's permitted academic history quickly and reliably.

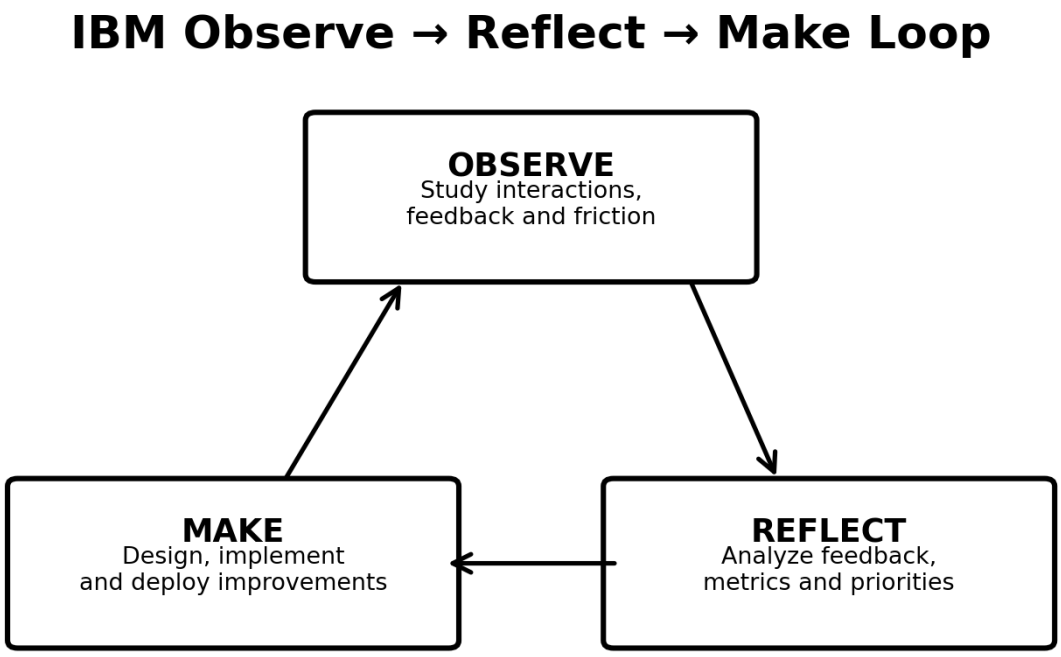
11.2 Restless Reinvention

The platform should not be considered finished after the first release. Usage patterns, support issues, feedback and operational metrics can identify opportunities for redesign and optimization.

11.3 Diverse Empowered Teams

The product benefits from collaboration among product/business representatives, UX contributors, software developers, QA, database specialists, academic-domain experts and IT/DevOps personnel.

11.4 IBM Loop



12. IBM Observe → Reflect → Make Loop

12.1 Observe

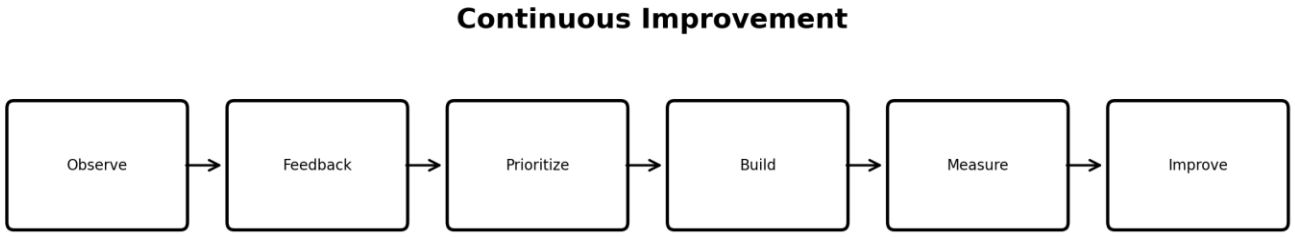
Collect evidence from user interactions, support requests, workflow delays, usability tests and operational metrics.

12.2 Reflect

Interpret the evidence, identify root causes and decide which problems matter most to user and business outcomes.

12.3 Make

Design, implement and release a focused improvement. Measure the result and return to observation.

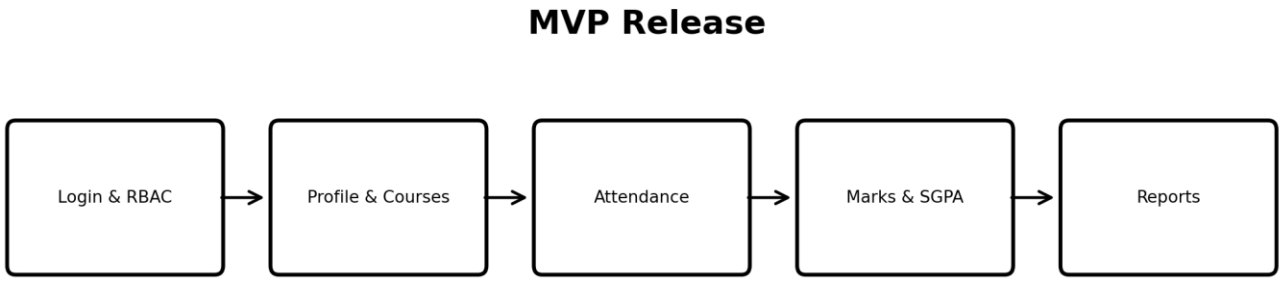


12.4 Example Improvement Cycle

Observation	Reflection	Action	Measure
Students miss registration deadlines.	Deadline information is fragmented.	Centralize deadline display and notifications.	Track notification engagement and late registrations.
Faculty spend too long entering attendance.	Manual entry is repetitive.	Provide quick attendance workflow.	Measure average attendance-entry time.
Result queries create peak load.	Many users access results simultaneously.	Optimize queries and caching strategy.	Measure response time during peak tests.

13. Minimum Viable Product (MVP)

The MVP is the smallest release that delivers meaningful value and validates the core product assumptions. It deliberately excludes features that are useful but not essential to proving the central product value.

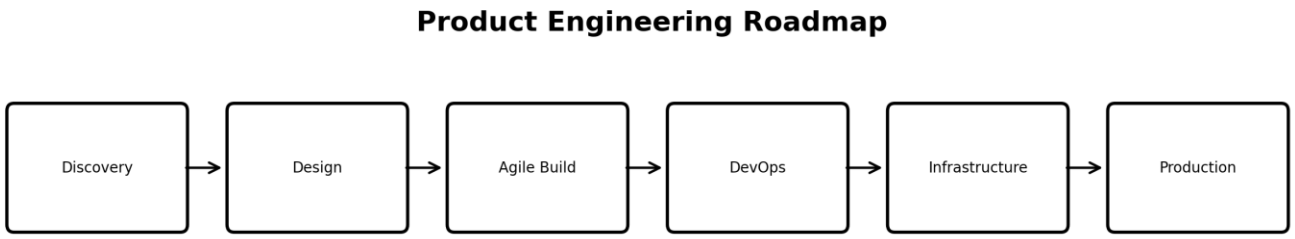


MVP Feature	Why It Matters	Priority
Secure Login & RBAC	Protects academic information and provides role-specific access.	Must Have
Student Profile & Course Registration	Establishes the core student academic record.	Must Have
Digital Attendance	Solves a major manual-record pain point.	Must Have
Marks & SGPA Calculation	Demonstrates automated academic processing.	Must Have
Standard Reports / Transcript	Provides a concrete administrative and student outcome.	Must Have

13.1 Deferred Features

- AI-based performance predictions.
- Hostel management.
- Fee/payment gateway.
- Alumni portal.
- Dedicated native mobile application.
- Advanced third-party integrations.

14. Product Roadmap

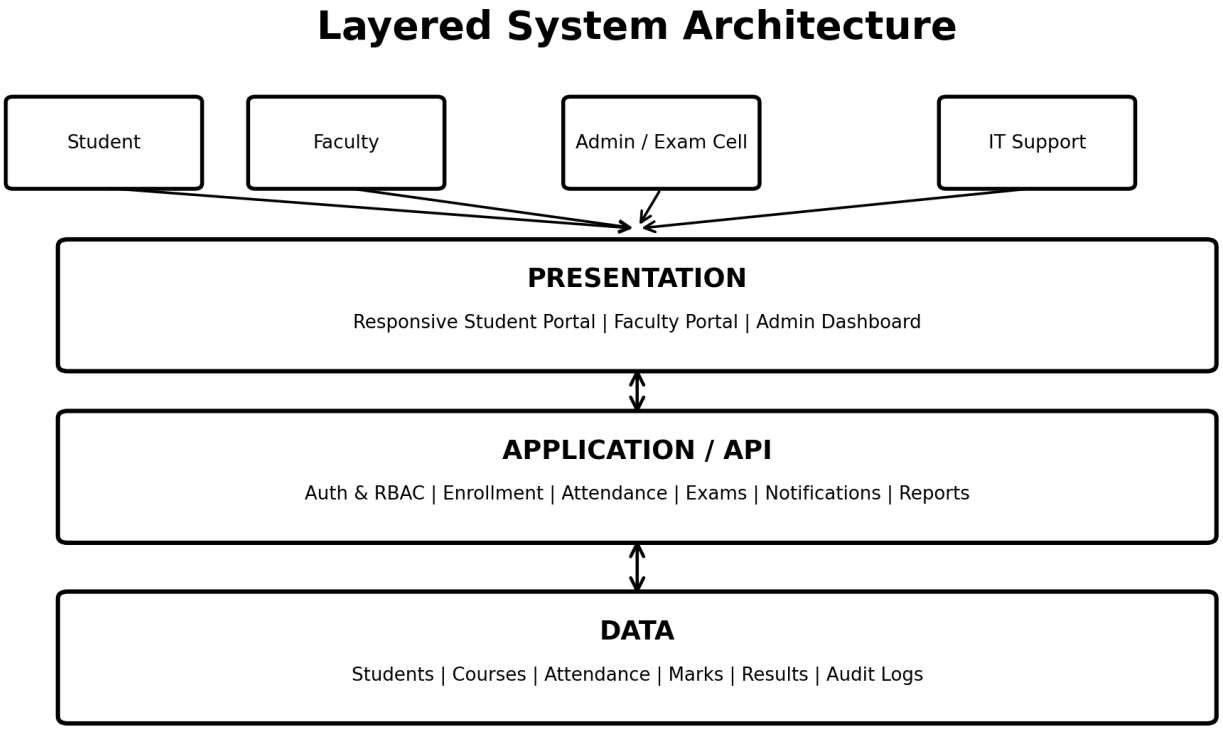


Phase	Focus	Key Outputs
Phase 1	Domain Discovery	Business problem, stakeholder map, personas, process understanding.
Phase 2	Design & Requirements	Design Thinking, BRD, FRS/NFRS, prototypes and acceptance criteria.
Phase 3	Agile Core Build	Authentication, profiles, courses, attendance, marks and reports.
Phase 4	DevOps	Version control, automated testing, CI/CD, containerization.
Phase 5	Infrastructure	Cloud provisioning, configuration management and scalable deployment.
Phase 6	Production & Monitoring	Deployment, logging, metrics, dashboards and continuous improvement.

14.1 Agile Delivery Principle

The roadmap is directional rather than a promise of fixed dates. Sprint priorities should be adjusted using validated user feedback, risk, dependencies and product outcomes.

15. Initial System Architecture



15.1 Architecture Layers

Layer	Responsibilities
Presentation Layer	Responsive interfaces for students, faculty and administrators.
Application/API Layer	Authentication, RBAC, registration, attendance, examinations, notifications and reporting.
Data Layer	Structured academic records, marks, attendance, course data and audit information.

15.2 Architectural Principles

- Separation of concerns: UI, application logic and persistence remain logically distinct.
- Security by design: authorization is enforced at the service boundary, not only in the interface.
- Modularity: academic modules can evolve independently where practical.
- Observability: production operation should expose useful logs, metrics and health information.
- Scalability: architecture should support growth without requiring a complete redesign.

16. Project Charter

Charter Element	Definition
Project Name	Enterprise Student Record and Academic Management Platform
Purpose	Digitize and centralize academic record and management workflows.
Objective	Deliver an MVP covering identity, profiles/courses, attendance, marks and academic reporting.
Primary Stakeholders	Students, Faculty, Academic Administrators, Examination Cell and IT Support.
Scope	Academic record lifecycle from onboarding through reporting, within defined MVP boundaries.
Major Risks	Requirement ambiguity, data quality, security/privacy issues, adoption resistance and peak-load performance.
Success	Reliable academic records, reduced manual work, usable workflows and positive pilot validation.
Governance	Defined product/business ownership, technical ownership, review checkpoints and acceptance criteria.

16.1 Major Milestones

27. Business/domain understanding completed.
28. Stakeholders and personas validated.
29. BRD and FRS/NFRS completed.
30. Design Thinking prototype tested.
31. MVP scope approved.
32. Architecture and repository initialized.
33. Development and quality-validation cycles begin.

17. Repository and Git Structure

17.1 Repository Hierarchy

```
student-record-academic-platform/
├── .github/            # CI/CD workflows
├── app/                # Application source code
│   ├── modules/        # Students, Courses, Attendance, Marks
│   ├── database/       # Schemas, migrations and ORM models
│   └── tests/          # Unit and integration tests
├── docs/               # Requirements, architecture and design reports
├── docker/             # Dockerfiles and local container setup
├── kubernetes/         # Deployment/service/ingress manifests
├── terraform/          # Infrastructure provisioning
├── ansible/            # Configuration management
├── monitoring/         # Monitoring dashboard configuration
└── README.md           # Setup and onboarding instructions
```

17.2 Git Initialization

A standard repository provides a shared source of truth for the engineering team. A protected main branch, feature branches, pull requests and automated checks can reduce accidental regressions.

Example:

```
git init

git add .

git commit -m "Initial project structure"

git branch -M main
```

17.3 Engineering Quality Gates

- Code review before merging significant changes.
- Automated unit and integration tests.
- Static analysis and dependency checks where available.
- Build verification before deployment.
- Documented release and rollback procedures.

18. Product Discovery Documentation

Product discovery is the evidence-gathering stage that connects the business problem to product decisions. The team studies current workflows, identifies pain points, validates assumptions and translates findings into prioritized outcomes.

18.1 Discovery Questions

- Which academic tasks consume the most staff time?
- Where do students experience the greatest uncertainty or delay?
- Which records require the highest level of accuracy and authorization?
- Which workflows create duplicate data entry?
- What information is needed during peak examination periods?
- What would make users trust the digital system?

18.2 Discovery-to-Backlog Traceability

Finding	User Need	Product Response	Backlog Theme
Attendance is hard to track manually.	Students need timely visibility.	Attendance dashboard + alerts.	Attendance
Marks are entered repeatedly.	Faculty need efficient entry.	Bulk grade entry + validation.	Examinations
Reports require manual compilation.	Admins need consistent reporting.	Centralized report generation.	Reporting
Information is fragmented.	Users need one source of truth.	Centralized academic record.	Platform foundation

19. Risk, Quality and Success Measures

19.1 Product and Engineering Risks

Risk	Impact	Mitigation
Unclear institutional rules	High	Validate requirements with academic stakeholders before implementation.
Incorrect or inconsistent data	High	Validation, database constraints, audit trails and controlled imports.
Unauthorized record access	Very High	RBAC, secure authentication, least privilege and auditability.
Low user adoption	High	Prototype testing, training, usability improvements and pilot rollout.
Peak-load performance problems	Medium/High	Performance testing, query optimization and scalable architecture.
Scope expansion	High	Protect MVP boundaries and prioritize through a product backlog.

19.2 Quality Strategy

- Functional testing validates business rules and workflows.
- Usability testing validates whether users can complete important tasks.
- Security testing checks authentication, authorization and common vulnerabilities.
- Performance testing evaluates response under representative load.
- Regression testing protects previously validated functionality.

19.3 Key Product Metrics

Metric	Why It Matters
Registration completion rate	Measures whether students can complete a core task successfully.
Attendance-entry time	Measures faculty workflow efficiency.
Result publication turnaround	Measures administrative improvement.
Record error rate	Measures data quality.
Task success / usability feedback	Measures user experience.
Availability and response time	Measures technical reliability.

20. Conclusion

The Enterprise Student Record and Academic Management Platform demonstrates how a software product should be planned before large-scale implementation. The project combines business understanding, stakeholder analysis, user personas, functional and non-functional requirements, Design Thinking, IBM Design Thinking, MVP planning, roadmap definition, architecture and engineering practices.

The strongest lesson from the project is that technology should serve a validated user and business need. A successful academic platform is therefore not merely a database or collection of screens; it is an evolving product that connects people, processes and reliable information.

The proposed MVP establishes a focused starting point, while the roadmap and architecture provide a path toward DevOps automation, scalable infrastructure, monitoring and continuous product improvement.

Appendix A – Deliverables Checklist

- Business Problem Statement & Business Justification
- Product Vision & MVP Scope
- Stakeholder Analysis & User Personas
- Functional Requirements Specification
- Non-Functional Requirements Specification
- Design Thinking Report
- IBM Design Thinking Report
- IBM Observe → Reflect → Make Loop
- MVP Definition
- Product Roadmap
- System Architecture
- Project Charter
- Git/Repository Structure
- Product Discovery Documentation
- Risk & Quality Strategy

Appendix B – Visual Index

Figure	Purpose
Figure 1	Software Development Lifecycle
Figure 2	Design Thinking process
Figure 3	IBM Observe → Reflect → Make loop
Figure 4	Layered system architecture
Figure 5	Stakeholder relationship map
Figure 6	Academic lifecycle workflow
Figure 7	MVP release flow
Figure 8	Product engineering roadmap
Figure 9	Continuous improvement cycle